Decoy-State BB84 Quantum Key Distribution

Security-Aware Simulation and Key-Rate Estimation

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1 Abstract

We implement a practical decoy-state BB84 QKD simulation based on weak coherent pulses, threshold detection, and realistic channel imperfections. The pipeline includes state preparation, channel and detector modeling, sifting and test/data split, decoy-state parameter estimation, asymptotic key-rate calculation, and finite-key corrections with privacy amplification penalties. We present gains and QBER across decoy intensities and show the key rate as a function of channel loss.

2 Introduction

Quantum key distribution (QKD) allows two parties to establish a shared secret key with information-theoretic security. In practical implementations, weak coherent pulses (WCPs) replace ideal single-photon sources, enabling photon-number-splitting attacks unless countered by the decoy-state method. This project simulates the decoy-state BB84 protocol end-to-end and estimates key rates from observed statistics.

3 Protocol Overview

The simulated protocol follows standard decoy-state BB84 structure:

1. Alice generates random bits and bases and selects pulse intensities (signal, decoy, vacuum).
2. Coherent pulse photon numbers are sampled from a Poisson distribution with mean μ.
3. The channel introduces loss; Bob uses threshold detectors with dark counts.
4. Alice and Bob sift to matching bases and split rounds into test and data sets.
5. Decoy-state statistics estimate single-photon yield and error rate.
6. Privacy amplification and error correction determine the final key length.

4 Simulation Model

4.1 State Preparation

Each round samples a basis (Z/X) and an intensity label (signal, decoy, vacuum). Photon numbers follow a Poisson distribution:

$$P(n \mid \mu) = \frac{\mu^n e^{-\mu}}{n!}.$$

4.2 Channel and Detection

Channel loss is modeled in dB; overall transmittance is

$$\eta = 10^{-\frac{\text{loss}_{\text{dB}}}{10}} \cdot \eta_d,$$

where η_d is detector efficiency. Threshold detectors click with probability $1-(1-p_d)(1-\eta)^n$, where p_d is dark-count probability and n is photon number. Misalignment introduces a bit-flip error with probability e_d.

4.3 Sifting and Test/Data Split

Sifting keeps rounds where Alice and Bob choose the same basis and Bob registers a click. Sifted rounds are randomly split into test and data sets using a configurable test fraction.

5 Decoy-State Parameter Estimation

We estimate gains and QBER per intensity and bound the single-photon yield Y_1 and error rate e_1 with the standard two-decoy formulas. For finite keys, we apply Hoeffding bounds to obtain conservative intervals for gains and QBER.

6 Key-Rate Estimation

The asymptotic decoy-state key rate per pulse is computed as

$$R = p_1 Y_1 \left[1 - h(e_1)\right] - Q_\mu f_{\text{EC}} h(E_\mu),$$

where $p_1 = \mu e^{-\mu}$, Q_μ is signal gain, E_μ is signal QBER, f_{EC} is error-correction efficiency, and $h(\cdot)$ is binary entropy. Finite-key results include a privacy amplification penalty of $2\log_2(1/\varepsilon_{\text{PA}})$ bits.

7 Results

7.1 Gain and QBER by Intensity

Figure 1 shows the gains and QBER for signal, decoy, and vacuum intensities.

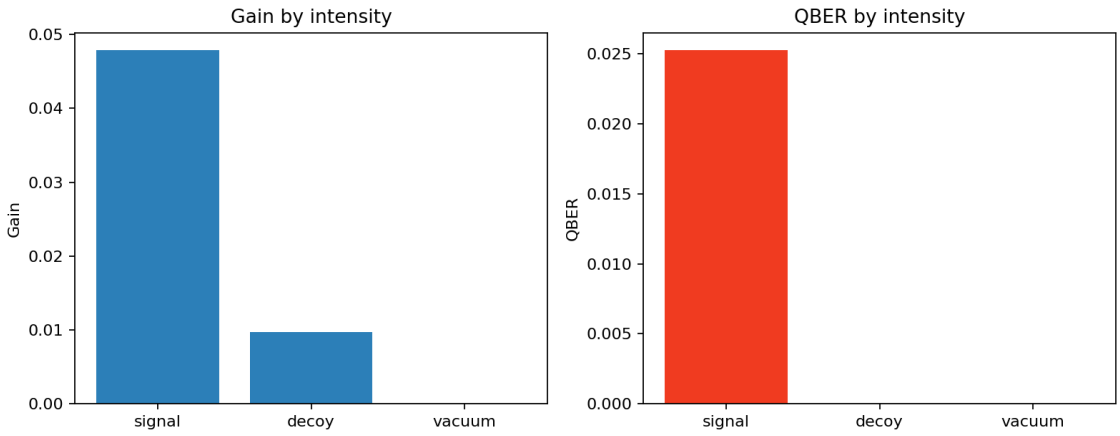


Figure 1: Gain and QBER by intensity.

7.2 Key Rate vs Channel Loss

Figure 2 shows the asymptotic key rate decreasing with channel loss.

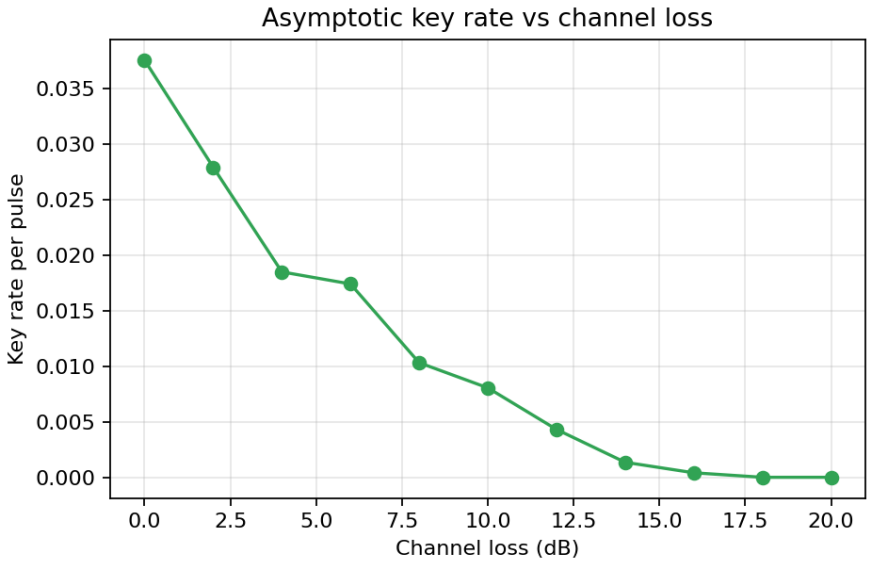


Figure 2: Asymptotic key rate per pulse vs channel loss.

8 Conclusion

The project demonstrates a full decoy-state BB84 pipeline and highlights the impact of realistic detector effects and finite-key corrections. The results are consistent with expected behavior: higher gains for signal intensities, lower gains for decoys, and a key rate that diminishes with channel loss. The finite-key bounds are conservative and can be improved with larger sample sizes or tighter concentration inequalities.

9 How to Run

- Simulation: `python -m src.main`
- Plots: `python -m src.plotting`